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Lab 6 Extra Datasets

Aim:To design and implement an Artificial Neural Network for classification problem and evaluate its learning performance.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
path = "/content/drive/MyDrive/ML datasets/Heart_Failure_Prediction_dataset.csv"
df = pd.read_csv(path)
df.head()
```

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	0
1	49	F	NAP	160	180	0	Normal	156	N	1.0	Flat	1
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	0
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	0

Next steps: Generate code with df New interactive sheet

```
print(df.columns)
```

```
Index(['Age', 'Sex', 'ChestPainType', 'RestingBP', 'Cholesterol', 'FastingBS',
       'RestingECG', 'MaxHR', 'ExerciseAngina', 'Oldpeak', 'ST_Slope',
       'HeartDisease'],
      dtype='object')
```

```
df.shape
```

```
(918, 12)
```

```
X = df.drop(columns=['HeartDisease'])
y = df['HeartDisease']
```

```
X = pd.get_dummies(X, drop_first=True)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = RandomForestClassifier()
model.fit(X_train, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier()
```

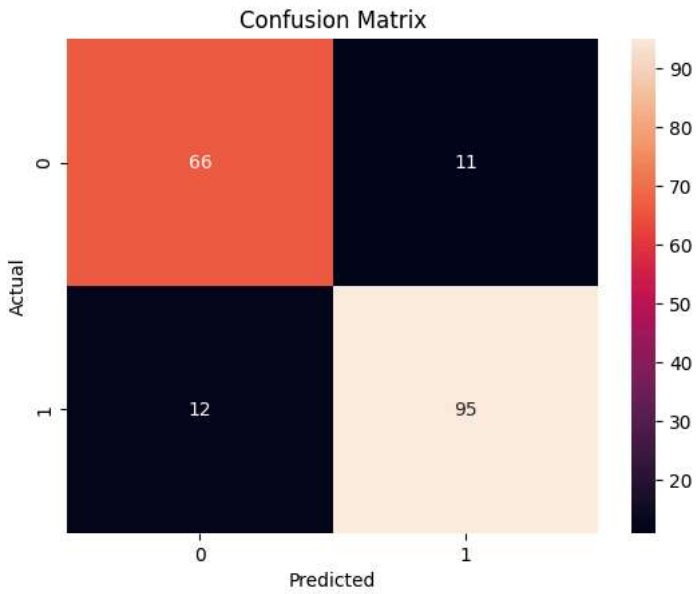
```
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.875
```

```
cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```




```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.85      0.86      0.85        77
           1       0.90      0.89      0.89       107

    accuracy                           0.88       184
   macro avg       0.87      0.87      0.87       184
weighted avg       0.88      0.88      0.88       184
```

```
from sklearn.linear_model import LogisticRegression

log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train, y_train)

y_pred_log = log_model.predict(X_test)

print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_log))
```

```
Logistic Regression Accuracy: 0.8532608695652174
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier()
dt_model.fit(X_train, y_train)
```

```
y_pred_dt = dt_model.predict(X_test)

print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
```

```
Decision Tree Accuracy: 0.8369565217391305
```

```
print("Random Forest:", accuracy_score(y_test, y_pred))
print("Logistic Regression:", accuracy_score(y_test, y_pred_log))
print("Decision Tree:", accuracy_score(y_test, y_pred_dt))
```

```
Random Forest: 0.875
Logistic Regression: 0.8532608695652174
Decision Tree: 0.8369565217391305
```

```
from sklearn.svm import SVC
svm = SVC(probability=True)
svm.fit(X_train, y_train)
y_pred_svm = svm.predict(X_test)
print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
```

```
SVM Accuracy: 0.6847826086956522
```

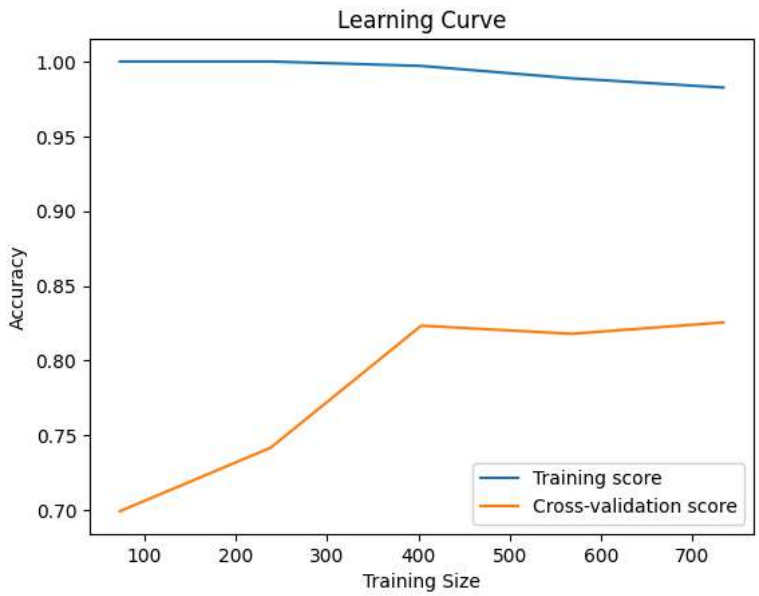
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [None, 10, 20]
}
grid = GridSearchCV(RandomForestClassifier(), param_grid, cv=5)
grid.fit(X_train, y_train)
print("Best Parameters:", grid.best_params_)
print("Best Accuracy:", grid.best_score_)
```

```
Best Parameters: {'max_depth': 10, 'n_estimators': 100}
Best Accuracy: 0.8732736930388594
```

```
train_acc = grid.score(X_train, y_train)
test_acc = grid.score(X_test, y_test)
print("Training Accuracy:", train_acc)
print("Testing Accuracy:", test_acc)
```

```
Training Accuracy: 0.9809264305177112
Testing Accuracy: 0.875
```

```
from sklearn.model_selection import learning_curve
train_sizes, train_scores, test_scores = learning_curve(
    grid.best_estimator_, X, y, cv=5
)
plt.plot(train_sizes, train_scores.mean(axis=1), label='Training score')
plt.plot(train_sizes, test_scores.mean(axis=1), label='Cross-validation score')
plt.xlabel("Training Size")
plt.ylabel("Accuracy")
plt.title("Learning Curve")
plt.legend(loc="best")
plt.show()
```




```
from sklearn.neighbors import KNeighborsClassifier
knn_model = KNeighborsClassifier()
knn_model.fit(X_train, y_train)
y_pred_knn = knn_model.predict(X_test)
results = {
    "KNN": accuracy_score(y_test, y_pred_knn),
    "SVM": accuracy_score(y_test, y_pred_svm),
    "Random Forest Tuned": accuracy_score(y_test, y_pred)
}
for model, acc in results.items():
    print(model, ":", acc)
```

```
KNN : 0.7065217391304348
SVM : 0.6847826086956522
Random Forest Tuned : 0.875
```